

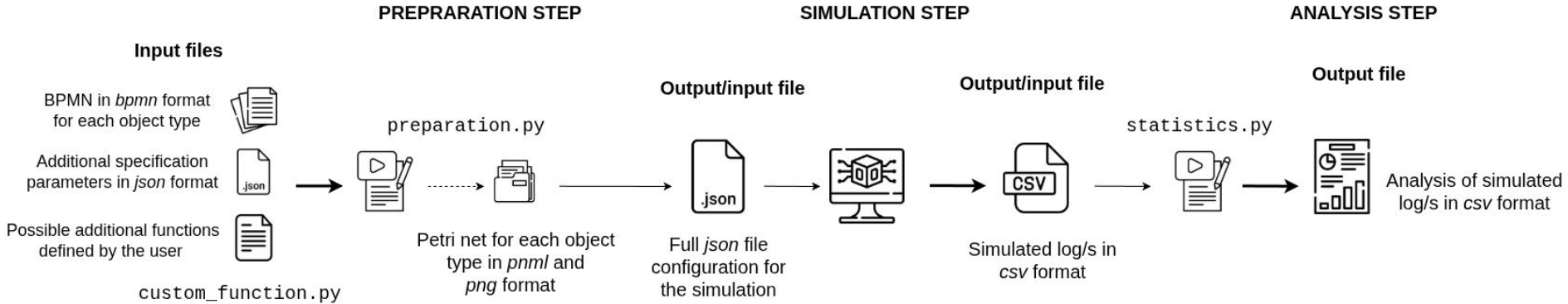


DORY Dynamic Object-centric Relations for sYnthetic simulation

DORY

- DORY is an **object-centric process simulator** designed to support multi-object processes.
- Implemented as a **Discrete Event Simulation** (DES) framework
 - **Object instances** can be **dynamically generated** and **removed**
 - **Relationships** between objects **evolve** according to the process behavior.

DORY



DORY



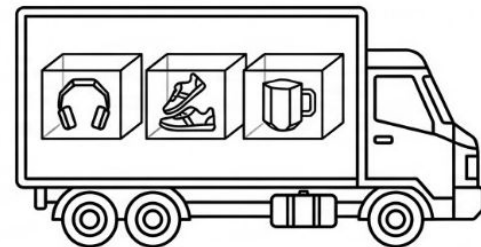
Order



User's home



Warehouse



Time

Delivery address

$[sum(price(I)) = 0]$

Place order

Check out

Pay CC

Payment

Close order

Order paid

Order created

Order updated

Truck capacity

Prepare

Create

N

1

Sync

ALL

1

Load

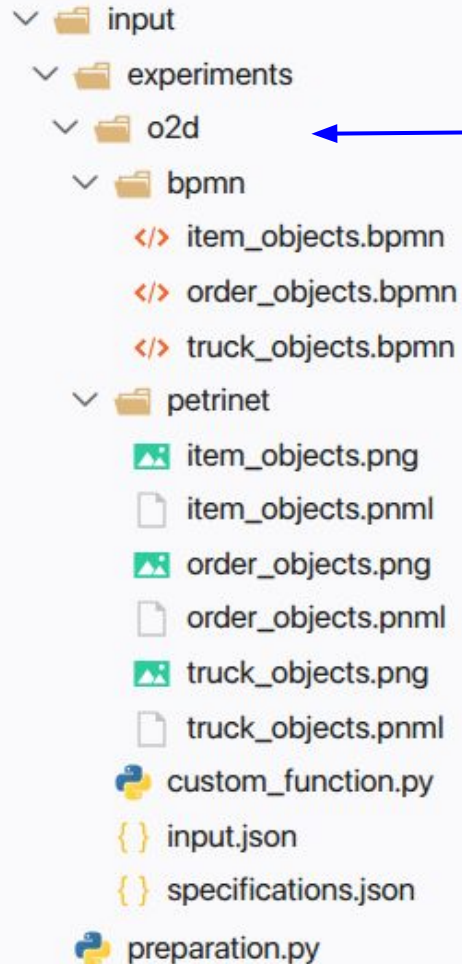
Ship

Go back



Truck

DORY Preparation step



1

2

```
python preparation.py o2d
```

Files automatically generated

4

input file for simulation

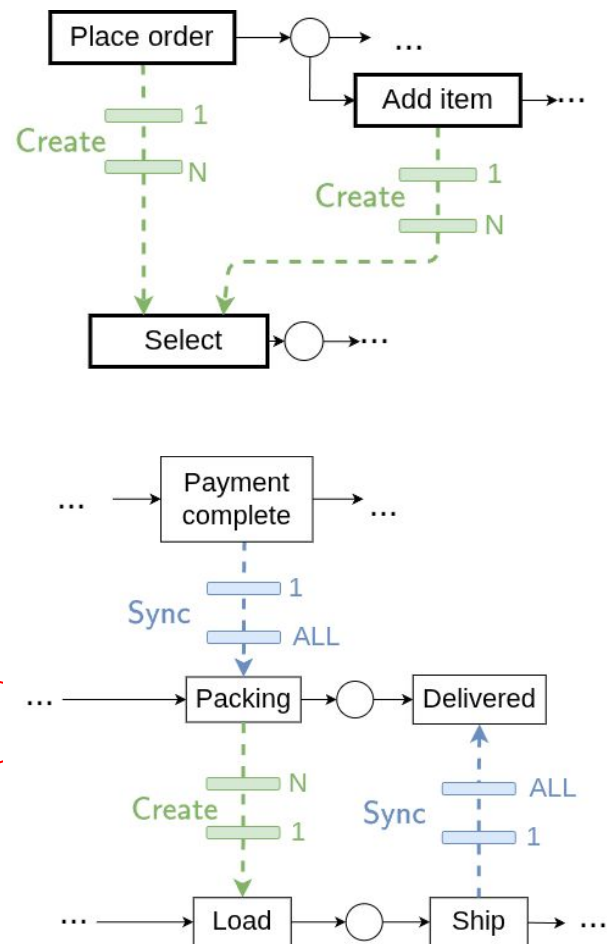
3

```

1 {
2   "start_simulation": "2026-01-01 09:00:00",
3   "objects": {
4     "order": {
5       "n_objects": 10,
6       "generator_by": [],
7       "generate": ["item"],
8       "task_generator": {
9         "Place Order": {"obj": "item", "name": "uniform", "parameters": {"low": 2, "high": 3}},
10        "Add Item": {"obj": "item", "name": "uniform", "parameters": {"low": 1, "high": 1}}
11      },
12      "waiting_time": {
13        "Place Order": {"name": "uniform", "parameters": {"low": 10, "high": 10}}
14      },
15      "object_constraints": {
16        "Check out": {"obj": "item", "trans": ["Pick Item", "Remove Item"], "card": "All"}
17      },
18      "create_relationship": {},
19      "destroy_relationship": {}
20    },
21    "item": {
22      "generator_by": ["order"],
23      "generate": [],
24      "task_generator": [],
25      "object_constraints": {
26        "Packing": {"obj": "order", "trans": ["Payment Complete"], "card": "All"},
27        "Delivered": {"obj": "truck", "trans": ["Ship"], "card": "All"}
28      },
29      "create_relationship": {},
30      "destroy_relationship": {"Remove Item": "order"}
31    },
32    "truck": {
33      "n_objects": 2,
34      "object_constraints": {
35        "Load": {"obj": "item", "trans": ["Packing"], "card": ["CUSTOM"]}
36      },
37      "create_relationship": {"Load": "item"},
38      "destroy_relationship": {"Ship": "item"}
39    }
40  }
41 }

```

specifications.json



input.json

```

1  {
2    "start_simulation": "2026-01-01 09:00:00",
3    "objects": {
4      "item": {
5        "path_petrinet": "petrinet/item_objects.pnml",
6        "interTriggerTimer": {},
7        "processing_time": {
8          "Select": {"name": "uniform", "parameters": {"low": 1, "high": 5}},
9          "Remove Item": {"name": "uniform", "parameters": {"low": 1, "high": 5}},
10         "Pick Item": {"name": "uniform", "parameters": {"low": 1, "high": 5}},
11         "Packing": {"name": "uniform", "parameters": {"low": 1, "high": 5}},
12         "Delivered": {"name": "uniform", "parameters": {"low": 1, "high": 5}}
13       },
14       "waiting_time": {},
15       "resource": {
16         "Role item": {"resources": ["item1", "item2", "item3"], "calendar": {"days": [0, 1, 2, 3, 4, 5, 6], "hour_min": 0, "hour_max": 23}}
17       },
18       "resource table": {"Select": "Role item", "Remove Item": "Role item", "Pick Item": "Role item", "Packing": "Role item", "Delivered": "Role item"},
19       "probability": {
20         "Remove Item": "AUTO", "Pick Item": "AUTO"
21       },
22       "generator_by": ["order"],
23       "generate": [],
24       "task_generator": [],
25       "object_constraints": {
26         "Packing": {"obj": "order", "trans": ["Payment Complete"], "card": "All"},
27         "Delivered": {"obj": "truck", "trans": ["Ship"], "card": "All"}
28       },
29       "create_relationship": {},
30       "destroy_relationship": {"Remove Item": "order"}
31     },
32     "order": {
33       "n_objects": 4,
34       "path_petrinet": "petrinet/order_objects.pnml",
35       "interTriggerTimer": {"type": "distribution", "name": "uniform", "parameters": {"low": 1, "high": 10}},
36       "processing_time": {
37         "Place Order": {"name": "uniform", "parameters": {"low": 1, "high": 5}},
38         "Check out": {"name": "uniform", "parameters": {"low": 1, "high": 5}},
39         "Add Item": {"name": "uniform", "parameters": {"low": 1, "high": 5}},
40         "Pay BT": {"name": "uniform", "parameters": {"low": 1, "high": 5}},
41         "Close Order": {"name": "uniform", "parameters": {"low": 1, "high": 5}},
42         "Pay CC": {"name": "uniform", "parameters": {"low": 1, "high": 5}},
43         "Payment Complete": {"name": "uniform", "parameters": {"low": 1, "high": 5}}
44       },

```


DORY Simulation step



Simulation

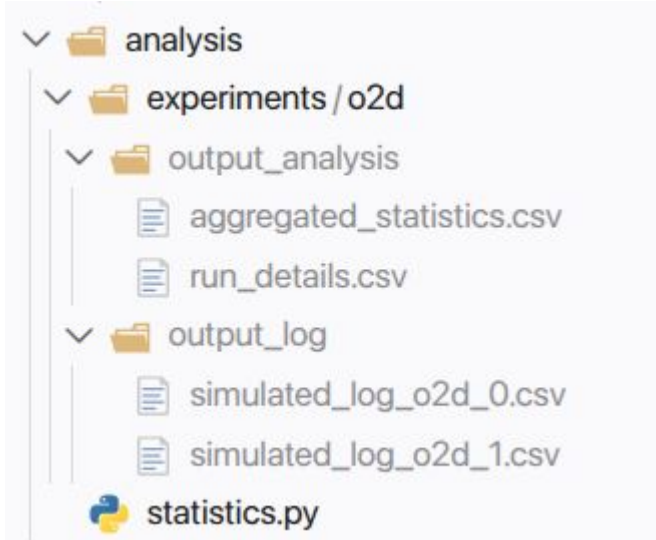
python run_simulation.py o2d

csv format

analysis > experiments > o2d > output_log > simulated_log_o2d_0.csv > data

```
1 obj_id,activity,enabled_time,start_time,end_time,role,resource,wip_wait,wip_start,wip_end,wip_activity,ro_total,ro_single,queue,prefix,relationships,attribute_object
2 truck_0,Prepare,2026-01-01 09:00:04,2026-01-01 09:00:04,2026-01-01 09:00:05,Role truck,truck1,1,2,1,"[0.0, 0.0, 0.33]",0.33,0,['Prepare'],{},{
3 order_0,Place Order,2026-01-01 09:00:05,2026-01-01 09:00:05,2026-01-01 09:00:07,Role order,order2,3,3,2,"[0.0, 0.67, 0.0]",0.67,0,['Place Order'],{"item_0_1", 'item_0_0'},{'City': 'Roma'}
4 order_1,Place Order,2026-01-01 09:00:04,2026-01-01 09:00:04,2026-01-01 09:00:07,Role order,order1,2,2,7,1,"[0.0, 0.33, 0.33]",0.33,0,['Place Order'],{"item_1_1", 'item_1_0'},{'City': 'Milano'}
5 item_0_1,Select,2026-01-01 09:00:07,2026-01-01 09:00:07,2026-01-01 09:00:09,Role item,item2,5,5,10,2,"[0.67, 0.33, 0.0]",0.67,0,['Select'],{'order_0'},{"Price": 883, 'City': 'Roma'}
6 item_0_0,Select,2026-01-01 09:00:07,2026-01-01 09:00:07,2026-01-01 09:00:10,Role item,item1,5,5,10,2,"[0.67, 0.33, 0.0]",0.67,0,['Select'],{'order_0'},{"Price": 263, 'City': 'Roma'}
7 item_1_1,Select,2026-01-01 09:00:07,2026-01-01 09:00:09,2026-01-01 09:00:10,Role item,item2,9,10,3,"[1.0, 1.0, 0.33]",1.0,0,['Select'],{'order_1'},{"Price": 395, 'City': 'Milano'}
8 order_2,Place Order,2026-01-01 09:00:07,2026-01-01 09:00:07,2026-01-01 09:00:10,Role order,order3,6,6,10,2,"[0.67, 0.67, 0.0]",0.67,0,['Place Order'],{"item_2_0", 'item_2_1'},{'City': 'Milano'}
9 item_1_0,Select,2026-01-01 09:00:07,2026-01-01 09:00:07,2026-01-01 09:00:11,Role item,item3,9,9,12,3,"[1.0, 0.67, 0.33]",1.0,0,['Select'],{'order_1'},{"Price": 602, 'City': 'Milano'}
10 truck_1,Prepare,2026-01-01 09:00:07,2026-01-01 09:00:07,2026-01-01 09:00:11,Role truck,truck2,7,7,12,1,"[0.67, 0.67, 0.33]",0.33,0,['Prepare'],{},{
11 order_3,Place Order,2026-01-01 09:00:08,2026-01-01 09:00:08,2026-01-01 09:00:11,Role order,order1,10,10,12,2,"[1.0, 1.0, 0.33]",1.0,0,['Place Order'],{"item_3_1", 'item_3_0'},{'City': 'Roma'}
12 item_0_0,Remove Item,2026-01-01 09:00:10,2026-01-01 09:00:10,2026-01-01 09:00:12,Role item,item2,10,10,14,2,"[1.0, 1.0, 0.33]",1.0,1,['Select', 'Remove Item'],{'order_0'},{"Price": 263, 'City': 'Roma'}
13 order_1,Add Item,2026-01-01 09:00:07,2026-01-01 09:00:07,2026-01-01 09:00:12,Role order,order2,7,9,13,1,"[0.67, 0.67, 0.33]",0.67,0,['Place Order', 'Add Item'],{"item_1_1", 'item_1_0', 'item_1_1'},{'City': 'Milano'}
14 item_1_1,Pick Item,2026-01-01 09:00:10,2026-01-01 09:00:11,2026-01-01 09:00:13,Role item,item3,10,12,14,1,"[1.0, 0.67, 0.33]",1.0,1,['Select', 'Pick Item'],{'order_1'},{"Price": 395, 'City': 'Milano'}
15 item_0_1,Remove Item,2026-01-01 09:00:09,2026-01-01 09:00:10,2026-01-01 09:00:13,Role item,item1,10,10,14,1,"[1.0, 1.0, 0.33]",1.0,1,['Select', 'Remove Item'],{'order_0'},{"Price": 883, 'City': 'Roma'}
16 item_2_0,Select,2026-01-01 09:00:10,2026-01-01 09:00:12,2026-01-01 09:00:15,Role item,item2,12,13,13,1,"[1.0, 0.67, 0.0]",1.0,1,['Select'],{'order_2'},{"Price": 312, 'City': 'Milano'}
17 order_1,Add Item,2026-01-01 09:00:12,2026-01-01 09:00:12,2026-01-01 09:00:15,Role order,order1,13,14,13,2,"[1.0, 0.67, 0.0]",0.67,0,['Place Order', 'Add Item'],{"item_1_1", 'item_1_0', 'item_1_1'},{'City': 'Milano'}
18 item_1_0,Pick Item,2026-01-01 09:00:11,2026-01-01 09:00:13,2026-01-01 09:00:15,Role item,item1,12,13,14,1,"[1.0, 1.0, 0.0]",1.0,3,['Select', 'Pick Item'],{'order_1'},{"Price": 602, 'City': 'Milano'}
19 order_0,Check out,2026-01-01 09:00:13,2026-01-01 09:00:13,2026-01-01 09:00:16,Role order,order2,13,13,14,1,"[1.0, 1.0, 0.0]",1.0,0,['Place Order', 'Check out'],{},{'City': 'Roma'}
20 order_3,Add Item,2026-01-01 09:00:11,2026-01-01 09:00:11,2026-01-01 09:00:16,Role order,order3,12,14,14,2,"[1.0, 0.67, 0.0]",0.67,0,['Place Order', 'Add Item'],{"item_3_2", 'item_3_1', 'item_3_1'},{'City': 'Milano'}
21 item_3_1,Select,2026-01-01 09:00:11,2026-01-01 09:00:15,2026-01-01 09:00:18,Role item,item1,14,14,15,3,"[1.0, 1.0, 0.0]",1.0,3,['Select'],{'order_3'},{"Price": 536, 'City': 'Roma'}
22 item_2_1,Select,2026-01-01 09:00:10,2026-01-01 09:00:13,2026-01-01 09:00:18,Role item,item3,12,14,15,2,"[1.0, 0.67, 0.0]",1.0,1,['Select'],{'order_2'},{"Price": 230, 'City': 'Milano'}
23 item_3_0,Select,2026-01-01 09:00:11,2026-01-01 09:00:15,2026-01-01 09:00:18,Role item,item2,14,13,15,2,"[1.0, 1.0, 0.0]",1.0,3,['Select'],{'order_3'},{"Price": 528, 'City': 'Roma'}
24 order_1,Add Item,2026-01-01 09:00:15,2026-01-01 09:00:15,2026-01-01 09:00:19,Role order,order1,13,14,15,2,"[1.0, 1.0, 0.0]",1.0,0,['Place Order', 'Add Item', 'Add Item', 'Add Item'],{"item_1_1", 'item_1_0', 'item_1_1'},{'City': 'Milano'}
25 item_2_0,Remove Item,2026-01-01 09:00:15,2026-01-01 09:00:18,2026-01-01 09:00:19,Role item,item3,13,15,16,1,"[1.0, 0.67, 0.0]",1.0,3,['Select', 'Remove Item'],{'order_2'},{"Price": 312, 'City': 'Milano'}
26 item_1_2,Select,2026-01-01 09:00:12,2026-01-01 09:00:18,2026-01-01 09:00:20,Role item,item1,14,15,15,3,"[1.0, 0.67, 0.0]",1.0,4,['Select'],{'order_1'},{"Price": 117, 'City': 'Milano'}
27 item_1_3,Select,2026-01-01 09:00:15,2026-01-01 09:00:18,2026-01-01 09:00:21,Role item,item2,14,15,15,2,"[1.0, 0.67, 0.0]",1.0,3,['Select'],{'order_1'},{"Price": 989, 'City': 'Milano'}
28 order_0,Pay BT,2026-01-01 09:00:16,2026-01-01 09:00:16,2026-01-01 09:00:21,Role order,order2,14,14,15,1,"[1.0, 1.0, 0.0]",1.0,0,['Place Order', 'Check out', 'Pay BT'],{},{'City': 'Roma'}
29 item_3_2,Select,2026-01-01 09:00:16,2026-01-01 09:00:19,2026-01-01 09:00:22,Role item,item3,15,15,15,3,"[1.0, 0.33, 0.0]",1.0,3,['Select'],{'order_3'},{"Price": 551, 'City': 'Roma'}
30 item_3_1,Pick Item,2026-01-01 09:00:18,2026-01-01 09:00:20,2026-01-01 09:00:23,Role item,item1,15,15,15,1,"[1.0, 0.33, 0.0]",1.0,4,['Select', 'Pick Item'],{'order_3'},{"Price": 536, 'City': 'Milano'}
31 item_2_1,Remove Item,2026-01-01 09:00:18,2026-01-01 09:00:21,2026-01-01 09:00:23,Role item,item2,15,15,15,1,"[1.0, 0.33, 0.0]",1.0,4,['Select', 'Remove Item'],{'order_2'},{"Price": 230, 'City': 'Milano'}
32 order_0,Payment Complete,2026-01-01 09:00:21,2026-01-01 09:00:21,2026-01-01 09:00:24,Role order,order3,15,15,14,1,"[1.0, 0.33, 0.0]",0.33,0,['Place Order', 'Check out', 'Pay BT'],{},{'City': 'Roma'}
33 item_1_4,Select,2026-01-01 09:00:19,2026-01-01 09:00:23,2026-01-01 09:00:26,Role item,item1,16,15,13,1,"[1.0, 0.33, 0.0]",1.0,4,['Select'],{'order_1'},{"Price": 349, 'City': 'Milano'}
```


DORY Analysis step



- Simulation results in csv format
- General statistical data
 - single run
 - aggregated data for multiple runs

```
python statistics.py o2d
```



object_centric_simulator

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10cda57 · 3 days ago

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.vscode

Added experiments folders

2 weeks ago



analysis

Fixed imports

last week



docs

documentation and custom file

3 days ago



imgs

Readme updated

2 months ago



input

documentation and custom file

3 days ago



reproducibility

Added InterTriggerTimer class

2 weeks ago



simulation

documentation and custom file

3 days ago

.gitignore

Added experiments folders

2 weeks ago

README.md

Quick fix readme

5 days ago

README



Simulating Dynamic Relationships in Object-Centric Processes

Installation guide

To execute this code, use **Python 3.10** and install the following main packages:

- scikit-learn==1.2.1
- scipy==1.11.2
- simpy==4.0.1

About

No description, website, or topics provided.

Readme

Activity

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Languages

Python 100.0%

Suggested workflows

Based on your tech stack

Dory

Overview

Video tutorial

Installation guide

Getting Started

Input files

Output files

Documentation

custom_function

object_class

inter_trigger_timer

parameters

process

run_simulation

role_simulator

example

utility

Examples



Dory: Dynamic Object-centric Relations for sYnthetic simulation

Object-centric process mining models real-world scenarios where multiple interdependent objects co-evolve during execution. Unlike traditional case-based approaches, object-centric event logs capture the full complexity of these interactions. However, existing simulation tools fail to capture the dynamic evolution of objects and their interactions, including their creation, modification, removal, and evolving relationships over time. DORY is an object-centric process simulator designed to address these limitations. The system supports multi- object processes where instances are dynamically generated, and removed throughout execution. By combining Petri net-based process definitions with configurable simulation parameters, DORY enables the generation of realistic object-centric event logs for process analysis.

